TEXORA · LECTURE SHEET

Weft Knitting Machines

The elements every weft machine shares — needles, sinkers, cams, feeders and take-down — arranged three ways: the circular cylinder (with or without a dial), the flat V-bed, and the small-diameter hosiery cylinder. Master the latch-needle cycle and the cam track, and every machine in this course is a variation on it.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 3 · LEVEL 3 TERM 1

1. Classification

FAMILY	NEEDLE BED	PRODUCT	TOPICS
Circular, cylinder only	Vertical needles in a rotating/static cylinder	Single-jersey tube	4
Circular, cylinder + dial	Vertical cylinder + horizontal dial needles at 90°	Rib, interlock tube	5, 6
Purl / links-links	Two beds in line, double-headed needles transfer	Purl panels	7
Flat (V-bed)	Two flat beds inclined as an inverted V	Panels, collars, whole garments	10
Small circular (hosiery)	Small cylinder, often with dial	Socks, stockings	11

Common measures: **gauge E** (needles/inch), **diameter** (circular, inches), **feeder (system) count**, and **speed factor** = diameter (m) × rpm for circular machines, a first check on whether a stated rpm is realistic for the size of machine.

2. The latch needle and its parts

The latch needle has: a **hook** at the top that receives yarn; a **latch** pivoted on a rivet just below the hook, whose spoon closes the hook; a **stem (shank)** down which the formed loop slides; and a **butt** near the lower end — the projection the cams push to raise and lower the needle. A longer **tail** below the butt may carry selection butts at different heights/positions. Needle quality is process quality: a bent

latch, burred hook or worn butt shows directly as dropped stitches, press-offs and vertical lines (needle lines) in the fabric.

3. The knitting cycle of the latch needle

1. **Rest / running-in:** the old loop sits in the closed hook; the needle is at its lowest working position and the fabric is held down by the sinkers.
2. **Clearing:** cams raise the needle by its butt. The old loop slides down the stem, pushes the latch open as it passes, and comes to rest below the open latch on the stem. The needle must rise far enough that the loop fully clears the latch — insufficient clearing is a prime cause of dropped stitches.
3. **Yarn feeding:** at the top of the rise the feeder presents new yarn across the open hook.
4. **Latch closing / loop formation:** the needle descends; the old loop, restrained by the sinker throat, slides up the stem, strikes the latch and closes it over the new yarn.
5. **Knock-over:** continuing down, the needle draws the new yarn through the old loop; the old loop is cast off over the closed hook (knocked over the sinker nib/knock-over edge) and becomes part of the fabric, while the new yarn forms the new loop in the hook.
6. **Loop length set:** how far the needle descends below the knock-over edge — set by the **stitch cam** — determines how much yarn is drawn, i.e. the stitch length.

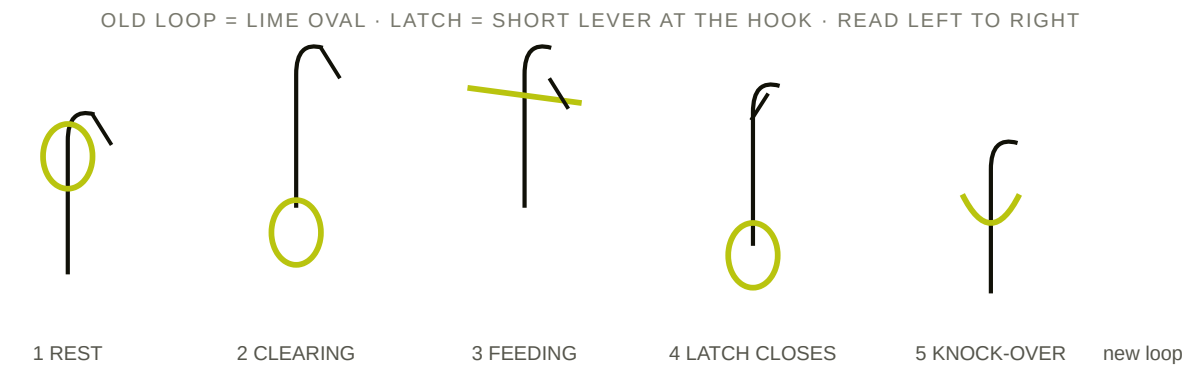


Figure 1 — The latch-needle cycle in five positions. The old loop itself opens the latch while clearing and closes it on the way down; the stitch cam's descent depth sets the new loop's length.

4. Sinkers, cams and the cam track

Sinkers are thin blades between the needles. Their throat holds the fabric down while needles rise to clear (without them the fabric would lift with the needles), their nib supports the old loop at knock-over, and the yarn is bent over their edge to form the sinker loop. In dial machines the dial has no sinkers — the dial loops are held by the cylinder fabric and dial sinker rings/tricks perform an equivalent holding role.

Cams are profiled blocks acting on needle butts. The working set at each feeder is: **clearing (raising) cam** — lifts butts to clearing height; **stitch cam** — adjustable in depth, pulls butts down to knock-over and sets stitch length; **guard/running cams** — keep butts in track between actions; plus **tuck and miss settings** — at tuck height the butt follows a lower track (loop not cleared); for miss, the needle is not raised at all. Needle selection for patterns is done by **jacks, selectors and (on electronic machines) piezoelectric/electromagnetic actuators** that push individual butts into or out of the cam track.

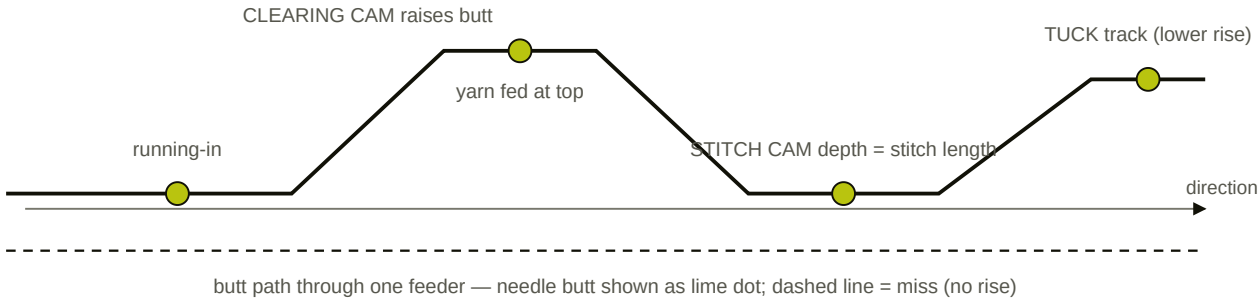


Figure 2 — Cam-track logic at one feeder. The same butt path, read as heights, is the knitting cycle: rise to clear, feed, descend past knock-over to the stitch-cam depth. Lower the clearing rise and the needle tucks; remove the rise and it misses.

5. Yarn feeding, fabric take-down and winding

Modern machines feed yarn under controlled positive feed: tape or belt-driven **positive feeders** (or storage feeders on jersey) deliver a set yarn length per revolution, which is what actually fixes stitch length uniformly across feeders; creels hold cones above or beside the machine, with tensioners, stop motions and knot catchers on each path. Below the needles, **take-down** (counter-tension, rollers, or motorised take-down) pulls the fabric tube away at a steady rate — take-down tension affects course spacing on the machine and must be even across the width. The tube is flattened and wound onto a roll, or plaited, by the **winder**; fabric is normally doffed by length, weight or time and inspected off-machine.

6. Machine data used throughout this course

$$\text{Total cylinder needles } N = \pi \times D\ (\text{inch}) \times E\ (\text{needles/inch})$$

Check: D = 30 in, E = 24 → N = π × 30 × 24 = 2261.9 → **2262 needles**. Every production and width calculation in Topics 4–9 and 12 starts from this number.

Glossary

Clearing — raising the needle until the old loop passes below the open latch. **Knock-over** — casting the old loop off over the closed hook. **Stitch cam** — the adjustable cam whose depth sets stitch length.

Sinker — blade between needles that holds fabric down and supports knock-over. **Feeder/system** — one complete yarn-feed + cam station; each feeder can form one course per revolution. **Speed factor** — π·D(m)·rpm-based surface-speed index, quoted as D(m) × rpm. **Positive feed** — metering a fixed yarn length per revolution to the needles.

Key points

• All weft machines run the same latch cycle: rest → clearing → feeding → latch closing → knock-over, with stitch-cam depth setting loop length.
• Cams + butts are the program; tuck and miss are simply lower and absent rises in the same track.
• Sinkers hold the fabric so needles can clear; positive feed holds stitch length equal across feeders; take-down must be steady and even.
• N = πDE is the needle count behind every later calculation.

Practice

1. List the five cycle positions in order and state, at each, where the old loop sits relative to hook and latch.
2. A needle rises but not fully to clearing height because of a worn clearing cam. Predict the fault pattern and its appearance in the fabric.
3. Calculate N for a 34-inch, E28 cylinder, and for an 18-inch, E28 cylinder (hosiery size).
4. Why does a dial (rib) machine manage without sinkers on the dial, and what suffers if dial/cylinder timing slips?
5. Stitch length increases although the stitch cam was not touched. Give three machine-side causes.

Answer key

1. Rest: old loop in the closed hook. Clearing: loop slides down the stem, opens the latch, rests below it. Feeding: loop below the open latch, new yarn laid in the hook. Latch closing: descending needle lets the held loop ride up and shut the latch over the new yarn. Knock-over: old loop cast off over the hook; new loop formed in the hook.
2. The old loop fails to pass below the latch on some needles, so the new yarn is trapped with it — dropped stitches and press-offs, appearing as holes/ladders repeating at the affected needles' wales and worsening at speed.
3. 34 in E28: π × 34 × 28 = 2990.8 → **2991 needles**. 18 in E28: π × 18 × 28 = 1583.4 → **1583 needles**.
4. The cylinder fabric and the intermeshing of the two beds hold the dial loops; dial sinker arrangements assist. If timing slips, cylinder and dial needles collide or loops are drawn unevenly — broken needles, holes and distorted rib wales.

5. Worn stitch-cam face or loose cam setting drifting down; positive-feed tape/belt slipping so more yarn is delivered per revolution; reduced yarn input tension or a feed-wheel diameter change after maintenance — any three consistent causes (also: take-down slack letting loops draw longer on the machine).
